

Data Warehousing and Data Mining

Module I -Introduction To Data Warehousing (6 hours)

- **Data preprocessing- Data cleaning – Data integration – Transformation-Data Reduction**
- **-Data Warehouse Characteristics- A multidimensional Data Model – OLAP Operations- Types of OLAP server-**
- **Data Warehouse Architecture and Implementation.**

Introduction

- Today's real-world databases are highly **susceptible to noisy, missing, and inconsistent data** due to their typically huge size and their likely origin from multiple, heterogeneous sources.
- Low-quality data will lead to **low-quality mining results**.
- How can the data be **preprocessed so as to improve the efficiency and ease of the mining process?**

What is a Data Warehouse?

- Defined in many different ways, but not rigorously.
 - A decision support database that is maintained **separately** from the organization's operational database
 - Support **information processing** by providing a solid platform of consolidated, historical data for analysis.
- “A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process.” —W. H. Inmon
- Data warehousing:
 - The process of constructing and using data warehouses

Data Warehouse—Subject-Oriented

- Organized around major subjects, such as **customer, product, sales**
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide **a simple and concise** view around particular subject issues by **excluding data that are not useful in the decision support process**

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources
 - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
 - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - E.g., Hotel price: currency, tax, breakfast covered, etc.
 - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
 - Operational database: current value data
 - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”

Data Warehouse—Nonvolatile

- A **physically separate store** of data transformed from the operational environment
- Operational **update of data does not occur** in the data warehouse environment
 - Does not require transaction processing, recovery, and concurrency control mechanisms
 - Requires only two operations in data accessing:
 - *initial loading of data* and *access of data*

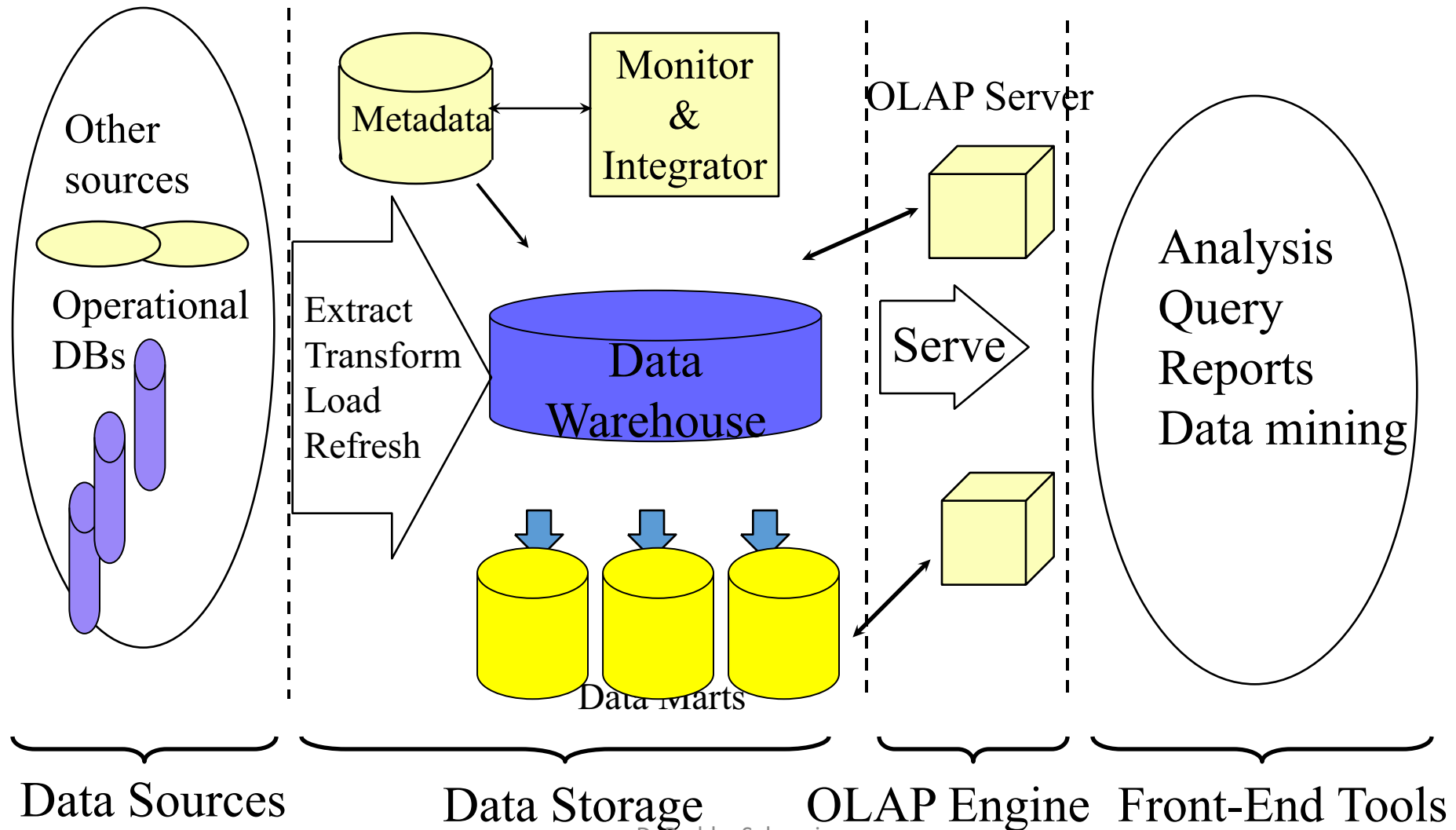
OLTP vs. OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Why a Separate Data Warehouse?

- High performance for both systems
 - DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
 - Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- Different functions and different data:
 - [missing data](#): Decision support requires historical data which operational DBs do not typically maintain
 - [data consolidation](#): DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
 - [data quality](#): different sources typically use inconsistent data representations, codes and formats which have to be reconciled
- Note: There are more and more systems which perform OLAP analysis directly on relational databases

Data Warehouse: A Multi-Tiered Architecture



Three Data Warehouse Models

- Enterprise warehouse
 - collects all of the information about subjects spanning the entire organization
- Data Mart
 - a subset of corporate-wide data that is of value to a specific groups of users. Its scope is confined to specific, selected groups, such as marketing data mart
 - Independent vs. dependent (directly from warehouse) data mart
- Virtual warehouse
 - A set of views over operational databases
 - Only some of the possible summary views may be materialized

Extraction, Transformation, and Loading (ETL)

- **Data extraction**
 - get data from multiple, heterogeneous, and external sources
- **Data cleaning**
 - detect errors in the data and rectify them when possible
- **Data transformation**
 - convert data from legacy or host format to warehouse format
- **Load**
 - sort, summarize, consolidate, compute views, check integrity, and build indices and partitions
- **Refresh**
 - propagate the updates from the data sources to the warehouse

Data Preprocessing

- There are several data preprocessing techniques.
 - Data cleaning can be applied to remove noise and correct inconsistencies in data.
 - Data integration merges data from multiple sources into a coherent data store such as a data warehouse.
 - Data reduction can reduce data size by, for instance, aggregating, eliminating redundant features, or clustering.
 - Data transformations (e.g., normalization) may be applied, where data are scaled to fall within a smaller range like 0.0 to 1.0.

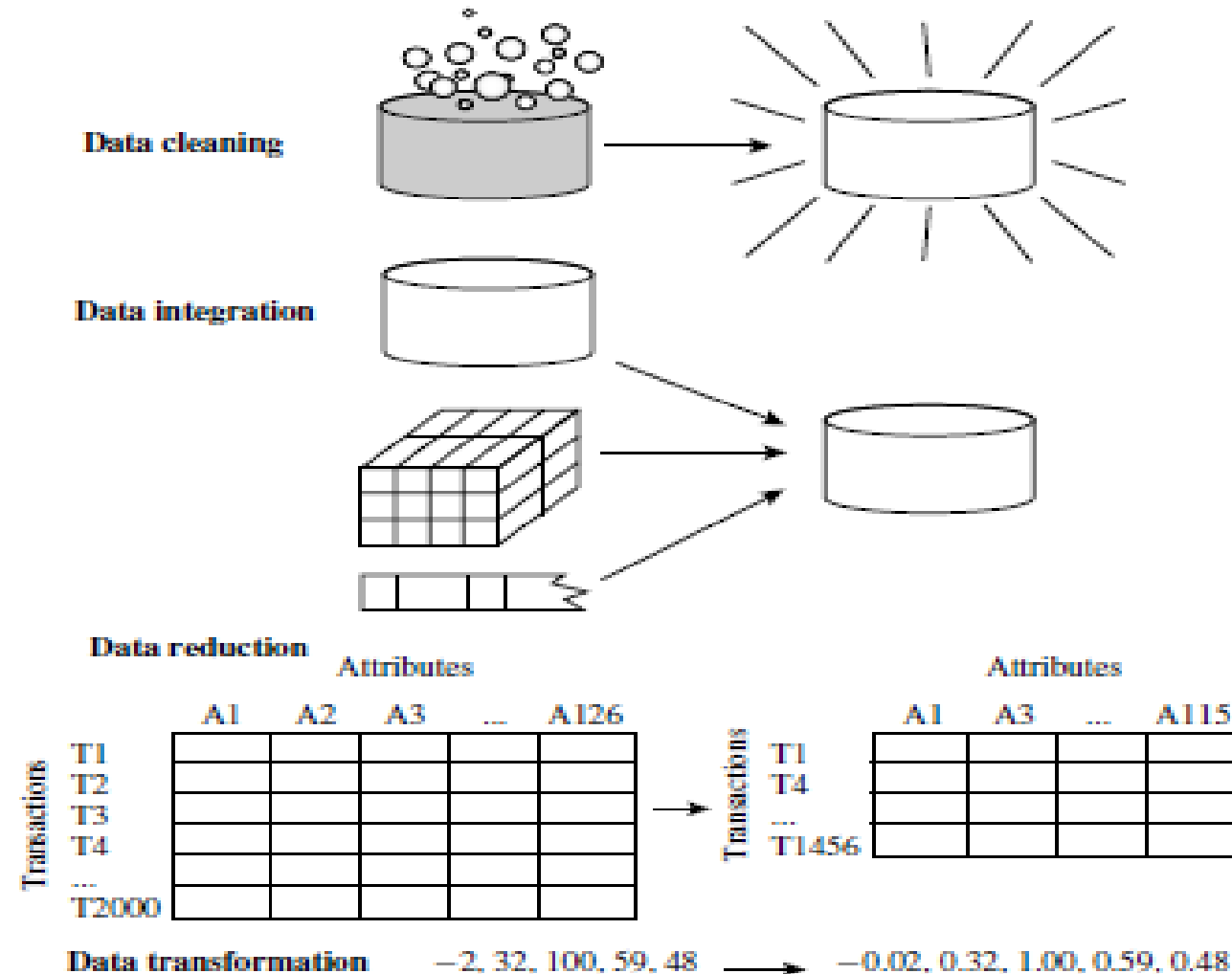
Data Quality: Why Preprocess the Data?

- Measures for data quality: A multidimensional view
 - **Factors comprising data quality**
 - **Accuracy**: correct or wrong, accurate or not
 - **Completeness**: not recorded, unavailable, ...
 - **Consistency**: some modified but some not, dangling, ...
 - **Timeliness**: timely update?
 - **Believability**: how trustable the data are correct?
 - **Interpretability**: how easily the data can be understood?

Major Tasks in Data Preprocessing

- **Data cleaning**
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
 - Integration of multiple databases, data cubes, or files
- **Data reduction**
 - Dimensionality reduction
 - Numerosity reduction
 - Data compression
- **Data transformation and data discretization**
 - Normalization
 - Concept hierarchy generation

Forms of data preprocessing.



Data preprocessing

Data preprocessing is a data mining technique that involves transforming raw data into an understandable format.

Real-world data is often incomplete, inconsistent, lacking in certain behaviors or trends, and is likely to contain many errors.

Data preprocessing refers to the set of techniques implemented on the databases to remove noisy, missing, and inconsistent data.

Task:

Data Cleaning

Data Integration

Data Transformation

Data Reduction

condt

Data Quality: Why Preprocess the Data?

No quality
data

age	income	student	buys_computer
22	high	yes	yes
40	high	no	yes
-22	medium		?
21		#e	no
12/03/2001	low	yes	
22/06/2000	low	Y	yes
34	medium	no	no

No quality
mining results



Data Pre-processing

- It is done to improve the quality of data in data warehouse
- **Data cleaning**
 - Process of removal of incorrect, incomplete ,inaccurate data (Noisy data)
 - replacing missing data/ Filling the missing value
 - Remove noisy data by using smoothing
 - Identify outliers data / dissimilar

Missind Data

- Ways to handle the missing data during cleaning:
- Simply delete row
- Manual Entry of missing data /Using “Most probable value “
/global contant /”unknown”/”N/A”

ID	Name	State	Degree
1	Aa	TN	UG
2	Bb	TN	UG
3	Cc		UG
4	Dd	TN	
5	Ee	TN	PG
6	Ff	TN	UG
7	Gg	AP	UG

• Find the mean, median for the following list of values:

13, 18, 13, 14, 13, 16, 14, 21, 13

The mean is the usual average, so:

$$(13 + 18 + 13 + 14 + 13 + 16 + 14 + 21 + 13) \div 9 = 15$$

The median is the middle value, so I'll have to rewrite the list in order:

13, 13, 13, 13, 14, 14, 16, 18, 21

There are nine numbers in the list, so the middle one will be the $(9 + 1) \div 2 = 5$ th number:

13, 13, 13, 13, 14, 14, 16, 18, 21

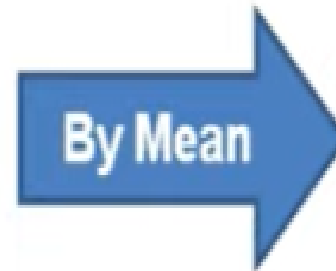
So the median is 14.

For Ungrouped Data (Simple Lists)

- **Mean:** $\frac{\text{Sum of all values}}{\text{Number of values}}$
 - *Example:* For {2, 3, 4, 5}, Mean = $(2+3+4+5) / 4 = 3.5$.
- **Median:**
 - Order the data. If n (count) is odd, it's the middle value (e.g., $(n+1)/2$ th value).
 - If n is even, it's the average of the two middle values (e.g., $n/2$ th and $(n/2 + 1)$ th values).
 - *Example ($n=4$):* {2, 3, 4, 5, 6} $\rightarrow (4+5)/2 = 4.5$.
- **Mode:** The value that appears most often.
 - *Example:* In {2, 3, 3, 4, 5}, the mode is 3. 

- Using attribute “ MEAN”
- Using Attribute”Medium” Using Attribute “Mode”

NO	A	B	C
1	6	6	4
2		8	2
3			2
4	4		4
5	6	4	
6	2	6	
Average	3	4	2



NO	A	B	C
1	6	6	4
2	3	8	2
3	3	4	2
4	4	4	6
5	6	4	2
6	2	6	2

- Noisy data: It is a random error / variance in measured variable.
- Incorrect data may due to
 - Data entry problems
 - Data transmission problems
 - Faulty data collection instrument

(1) Equal Frequency partition / Equal depth

Given: depth

(2) Equal width partition

(1) Equal Frequency partition / Equal depth

- Partition the data into Equidepth bins of depth 4:
- Data: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

- Bin 1: 5, 10, 11, 13
- Bin 2: 15, 35, 50, 55
- Bin 3: 72, 92, 204, 215

Smoothing Methods:

- Smoothing by Mean
- Smoothing by Median
- Smoothing by Boundary

- Noisy Data:

Binning:

Data:10,2,19,18,20,18,25,28,22

Sort:2,10,18,18,19,20,22,25,28

2,10,18

18,19,20

22,25,28

Bin Size =3

condt

- Smoothing by BIN Mean
- Value of bin is replaced by mean Value(Average)

$$2, 10, 18 \text{-----} 2+10+18=30/3=10$$

$$18, 19, 20 \text{-----} 18+19+20=57/3=19$$

$$22, 25, 28 \text{-----} 22+25+28=75/3=25$$

10,10,10

19,19,19

25,25,25

Condt

2,10,18

18,19,20

22,25,28

- Smoothing by Medians
- $\text{Odd} = n+1/2$
- $\text{Even} = \text{add two middle values}/2$

10,10,10

19,19,19

25,25,25

Condt

- Smoothing by Bin Boundaries:
- Two Min and Max value

2,10,18

18,19,20

22,25,28

- 2, 2, 18
- 18 18 20
- 22 22 28

Condt

- Example:

1. 4,8,9,15,21,21,24,25,26,28,29,34 depth =4
2. 4,8,15,21,21,24,25,28,34 bin=3
3. 4,8,9,15,21,21,24,25,26,28,29,34 DEPTH=3

Binning Methods for Data Smoothing

- * Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
- * Partition into bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
- * Smoothing by bin means:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
- * Smoothing by bin boundaries:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34

Example:-

• **Data:- 4, 8, 15, 21, 21, 24, 25, 28, 34**

1. Partition into equidepth

Bin1:- 4, 8, 15

Bin2:- 21, 21, 24

Bin3:- 25, 28, 34

2. Smooth by bin

Bin1:- 9, 9, 9

Bin2:- 22, 22, 22

Bin3:- 29, 29, 39

Example:-

3. Smooth by Boundaries

Bin1:- 4, 4, 15

Bin2:- 21, 21, 24

Bin3:- 25, 25, 34

Condt

(2) Equal width partition

$W = \text{Max-Min/bin size}$

$D = \{ 10, 15, 16, 18, 20, 30, 35, 42, 48, 50, 52, 55 \}$ bin size=3

$$W = 55 - 10 / 3 = 45 / 3 = 15$$

$$\text{Bin 1} = [10, \text{MIN} + W - 1] \text{ ----- } [10, 10 + 15 - 1] = [10, 24]$$

$$\text{Bin 2} = [\text{MIN} + W, \text{MIN} + 2W - 1] \text{ ----- } [10 + 15, 10 + 2 * 15 - 1] = [25, 39]$$

$$\text{Bin 3} = [\text{MIN} + 2w, \text{MAX}] \text{ ----- } [10 + 2 * 15, 55] = [40, 55]$$

Condt

- Bin1 :[10, 15,16,18,20] $(10+15+16+18+20/5=79/5=15.8)$
- Bin2 :[30,35] $(30+35/2=32.5)$
- Bin3:[42,48,50,52,55] $(42+48+50+52+55/5=247/5=49.4)$

- Smoothing by Mean
- Bin1 :[15.8, 15.8,15.8,15.8,15.8]
- Bin2 :[32.5,32.5]
- Bin3:[49.4,49.4,49.4,49.4,49.4]

Condt

- Smoothing by **Median**
- Bin1 : [10, 15, **16**, 18, 20] = [16, 16, 16, 16, 16]
- Bin2 : [**30, 35**] = [32.5, 32.5] ($30+35/2=32.5$)
- Bin3: [42, 48, **50**, 52, 55] = [50, 50, 50, 50, 50]
- Smoothing by **boundaries**
- Bin1 : [10, 15, 16, 18, 20] = [10, 10, 20, 20, 20] (REPLACE WITH MIN DIFFERENCE)
- Bin2 : [30, 35] = [32.5, 32.5] = [30, 35]
- Bin3: [42, 48, 50, 52, 55] = [42, 42, 55, 55, 55]

Data Integration

- It is preprocessing method that involves merging of data from different sources in order to form a data store like data warehouse.

Issues in Data Integration:

1. Schema integration and object matching
2. Redundancy – unwanted attributes (Derived from other attributes)
3. Detection and resolution of data value conflicts(different representation, different scales...

Data transformation

- It is data pre-processing technique that transform the data into alternative forms appropriate for mining .

Techniques:

- Smoothing-Removing noise from data[binning by mean, medium, boundary, regression, clustering]
- Normalization- attribute's values are normalized by scaling their values . So that they fall in specified range

- Min – Max Normalization:
- Z-score normalization:
- Decimal Normalization
- **Min – Max Normalization:**
- v' - It is the new value of each attribute data.
- v - It is the old value of each attribute data

new_max_A , new_min_A is the max and min value within the range

M

[n

$$v' = \frac{v - \min_A}{\max_A - \min_A} (new_max_A - new_min_A) + new_min_A$$

condt

marks

8

10

15

20

Min:

The minimum value of the given attribute. Here Min is **8**

Max:

The maximum value of the given attribute. Here Max is **20**

V: V is the respective value of the attribute. For example here V1=8, V2=10, V3=15, and V4=20

newMax:

1

newMin:

0

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

For Marks as 8:

$$\text{MinMax} = \frac{(V - \text{Min marks})}{\text{Max o marks} - \text{Min marks}} (\text{newMax} - \text{newMin}) + \text{newMin}$$

$$\text{MinMax} = \frac{(8 - 8)}{20 - 8} * (1 - 0) + 0$$

$$\text{MinMax} = \frac{(0)}{12} * 1$$

$$\text{MinMax} = 0$$

For Marks as 10:

$$\text{MinMax} = \frac{(10 - 8)}{20 - 8} * (1 - 0) + 0$$

$$\text{MinMax} = \frac{(2)}{12} * 1$$

$$\text{MinMax} = 0.16$$

For Marks as 15:

$$MinMax = \frac{(15 - 8)}{20 - 8} * (1 - 0)^{\frac{1}{2}} + 0$$

$$MinMax = \frac{(7)}{12} * 1$$

$$MinMax = \mathbf{0.58}$$

For Marks as 20:

$$MinMax = \frac{(20 - 8)}{20 - 8} * (1 - 0)^{\frac{1}{2}} + 0$$

$$MinMax = \frac{(12)}{12} * 1$$

$$MinMax = 1$$

marks

8

10

15

20

marks after Min-Max normalization

0

0.16

0.58

1

Problem 2

- Suppose we have to normalize the following data set, **200, 300, 400, 600, 1000** to a new range [0, 1], then using min-max normalization
- $\text{min} = 200$, $\text{max} = 1000$, $\text{new_min} = 0$, $\text{new_max} = 1$

condt

$$v'_1 = \frac{200 - 200}{1000 - 200} (1 - 0) + 0 = 0$$

$$v'_2 = \frac{300 - 200}{1000 - 200} (1 - 0) + 0 = \frac{100}{800} = 0.125$$

$$v'_3 = \frac{400 - 200}{1000 - 200} (1 - 0) + 0 = \frac{200}{800} = 0.25$$

$$v'_4 = \frac{600 - 200}{1000 - 200} (1 - 0) + 0 = \frac{400}{800} = 0.5$$

$$v'_5 = \frac{1000 - 200}{1000 - 200} (1 - 0) + 0 = 1$$

condt

- Zero- Score Normalization

$$\text{New value} = (x - \mu) / \sigma$$

x : Original value

μ : Mean of data

σ : Standard deviation of data

condt

**How to calculate Z-Score of the following data?
marks**

8

10

15

20

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum (\text{every individual value of marks} - \text{mean of marks})^2}{n}}$$

$$\text{Mean of marks} = 8 + 10 + 15 + 20 / 4 = 13.25$$

$$= \sqrt{\frac{(8 - 13.25)^2 + (10 - 13.25)^2 + (15 - 13.25)^2 + (20 - 13.25)^2}{4}}$$

$$= \sqrt{\frac{(-5.25)^2 + (-3.25)^2 + (1.75)^2 + (6.75)^2}{4}}$$

$$= \sqrt{\frac{27.56 + 10.56 + 3.06 + 45.56}{4}} = \sqrt{\frac{86.74}{4}} = \sqrt{21.6} = 4.6$$

- Mean = 13.25
- Standard deviation = 4.6

$$ZScore = \frac{x - \mu}{\sigma} = \frac{8 - 13.25}{4.6} = -1.14$$

$$ZScore = \frac{x - \mu}{\sigma} = \frac{10 - 13.25}{4.6} = -0.7$$

$$ZScore = \frac{x - \mu}{\sigma} = \frac{15 - 13.25}{4.6} = 0.3$$

$$ZScore = \frac{x - \mu}{\sigma} = \frac{20 - 13.25}{4.6} = 1.4$$

marks	marks after z-score normalization
8	-1.14
10	-0.7
15	0.3
20	1.4

If a value is exactly equal to the mean of all the values of the feature, it will be normalized to 0.

If it is below the mean, it will be a negative number, if it is above the mean it will be a positive number

marks

marks after z-score normalization

8

-1.14

10

-0.7

15

0.3

20

1.4

- we have to normalize the following data set
- **200, 300, 400, 600, 1000** then using Z-Score normalization

<i>original data</i>	200	300	400	600	1000
<i>z-score</i>	-1.06	-0.7	-0.35	0.35	1.78

Decimal Scaling Normalization

Formula:

Dataset: [135, 20, -98, 400, -255]

Step 1: $\max(|x|) = 400$

$$\max(|x|) = \max(|135|, |20|, |-98|, |400|, |-255|) = 400$$

Step 2: Find j such that $10^j > 400 \Rightarrow j = 3$

$$\frac{400}{10^j} < 1 \Rightarrow 10^j > 400 \Rightarrow j = 3 \quad (\text{since } 10^3 = 1000 > 400)$$

$$\max(|x|) = \max(|135|, |20|, |-98|, |400|, |-255|) = 400$$

$$\frac{400}{10^j} < 1 \Rightarrow 10^j > 400 \Rightarrow j = 3 \quad (\text{since } 10^3 = 1000 > 400)$$

Step 3: Normalize using $x_{\text{normalized}} = x / 1000$

ID	Original Value	Normalized Value
1	135	0.135
2	20	0.02
3	-98	-0.098
4	400	0.4
5	-255	-0.255

Decimal Scaling

- A value v of attribute A is can be normalized by the following formula
- Normalized value of attribute = $(v^i / 10^j)$

Example of Decimal scaling :

CGPA	Formula	CGPA Normalized after Decimal scaling
2	$2/10$	0.2
3	$3/10$	0.3

-10, 201, 301, -401, 501, 601, 701

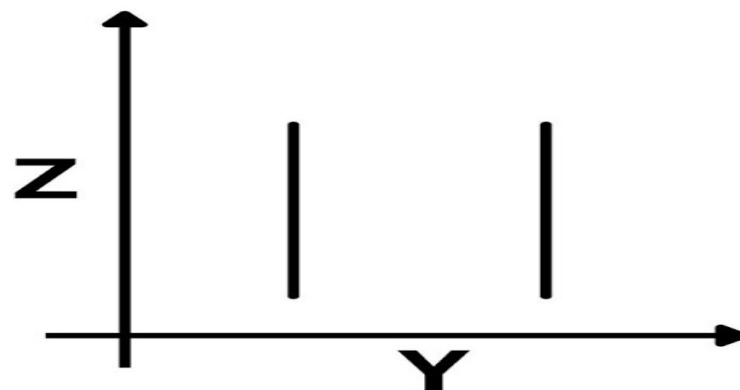
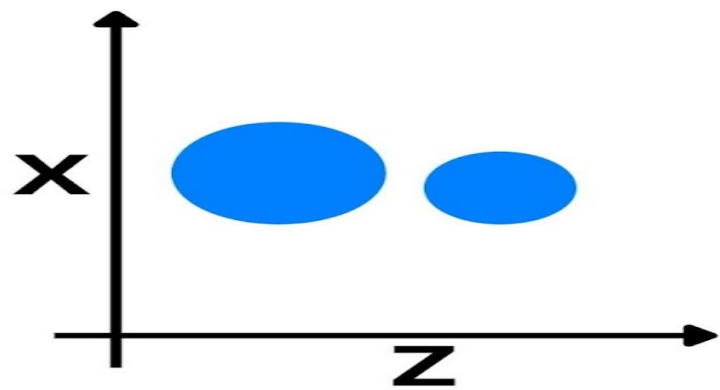
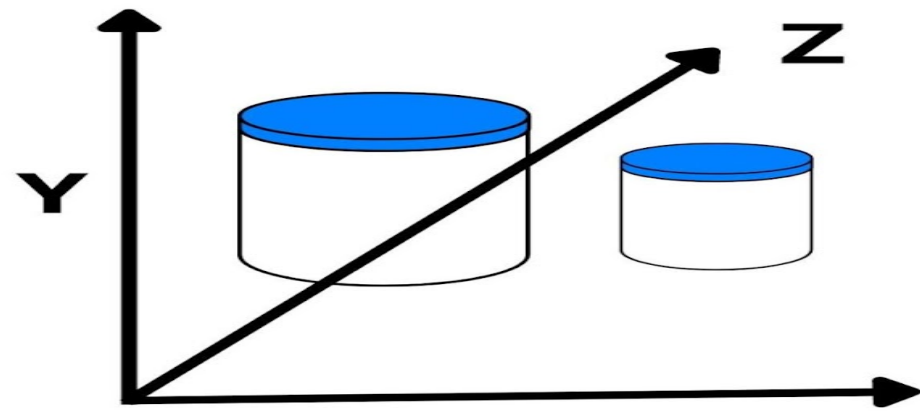
Maximum absolute value in given data(m): 701

So $j=3$ ie 1000

***The normalized data is: -0.01, 0.201, 0.301, -0.401,
0.501, 0.601, 0.701***

Data Reduction

- Reduced representation of dataset from the available dataset.
- Why data reduction? — A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set
- Easy visualization
- Improve computation time and space
- Remove irrelevant /noisy data
- -Integrity of the original data should be maintain even after reduction in data volume
- -It should produce same analytics result as on original data.



Dimensionality Reduction

- Dimensionality reduction[higher dimensional into lower dimensional dataset]
- Numerosity Reduction
- Data Compression

- Dimensionality reduction:

Types:

- Attribute selection/ feature selection(Subset selection)

- $\{x_1, x_2\}$ choose one out of two
- $\{x_1, x_2, x_3\}$ choose two out of three
 $\{x_1, x_2\}, \{x_2, x_3\}, \{x_3, x_1\} \dots$

- Every subset of original set of attributes can be considered .
- Find subset that has the best evaluation
- $\{x_1, x_2, \dots, x_D\}$
- Optimization Problem (finding best)
- Loss valuable information

To solve the characteristic equation for a 2x2 matrix:

For matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

1. **(A - λI):** $\begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}$

2. **Determinant:** $(a - \lambda)(d - \lambda) - bc$

3. **Equation:** $(a - \lambda)(d - \lambda) - bc = 0 \implies \lambda^2 - (a + d)\lambda + (ad - bc) = 0$ (This is $\lambda^2 - \text{Trace}(A)\lambda + \det(A) = 0$).

4. **Solve:** Use the quadratic formula: $\lambda = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$.

Principal component Analysis(PCA)

- Dimension reduction Techniques.
 1. Compute the mean vector
 2. Calculate covariance matrix
 3. Calculate the Eigen vectors & Eigen values of covariance matrix
 4. Choosing components and forming feature vector.
 5. Derive new data set.

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

- **Step1:**

$$\bar{X}_1 = \frac{1}{4}(4 + 8 + 13 + 7) = 8,$$

$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = 8.5.$$

Step 2: Calculation of the covariance matrix.

$$S = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

$$\begin{aligned}
&= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{1k} - \bar{X}_1) \\
&= \frac{1}{3} ((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2) \\
&= 14
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2) \\
&= \frac{1}{3} ((4-8)(11-8.5) + (8-8)(4-8.5) \\
&\quad + (13-8)(5-8.5) + (7-8)(14-8.5)) \\
&= -11
\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_2, X_1) &= \text{Cov}(X_1, X_2) \\ &= -11\end{aligned}$$

$$\begin{aligned}\text{Cov}(X_2, X_2) &= \frac{1}{N-1} \sum_{k=1}^N (X_{2k} - \bar{X}_2)(X_{2k} - \bar{X}_2) \\ &= \frac{1}{3} \left((11 - 8.5)^2 + (4 - 8.5)^2 + (5 - 8.5)^2 + (14 - 8.5)^2 \right) \\ &= 23\end{aligned}$$

$$\begin{aligned}
 S &= \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix} \\
 &= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}
 \end{aligned}$$

Step 3: Eigenvalues of the covariance matrix

$$\begin{aligned}
 0 &= \det(S - \lambda I) \\
 &= \begin{vmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{vmatrix} \\
 &= (14 - \lambda)(23 - \lambda) - (-11) \times (-11) \\
 &= \lambda^2 - 37\lambda + 201 \\
 &= \frac{1}{2}(37 \pm \sqrt{565}) \\
 &= 30.3849, 6.6151 \\
 &= \lambda_1, \lambda_2 \quad (\text{say})
 \end{aligned}$$

Condt (Eigenvectors and Unit Eigenvector)

Step 4: Computation of the eigenvectors

$$\begin{aligned}\begin{bmatrix} 0 \\ 0 \end{bmatrix} &= (S - \lambda I) U \\ &= \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \\ &= \begin{bmatrix} (14 - \lambda)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda)u_2 \end{bmatrix}\end{aligned}$$

condt

$$\begin{aligned}\|U_1\| &= \sqrt{11^2 + (14 - \lambda_1)^2} \\ &= \sqrt{11^2 + (14 - 30.3849)^2} \\ &= 19.7348\end{aligned}$$

$$\begin{aligned}&= \begin{bmatrix} 11/\|U_1\| \\ (14 - \lambda_1)/\|U_1\| \end{bmatrix} \\ &= \begin{bmatrix} 11/19.7348 \\ (14 - 30.3849)/19.7348 \end{bmatrix} \\ &= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}\end{aligned}$$

- **Step5 generate new dataset**

$$e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{21} - \bar{X}_2 \end{bmatrix}$$

$$= 0.5574(X_{11} - \bar{X}_1) - 0.8303(X_{21} - \bar{X}_2)$$

$$= 0.5574(4 - 8) - 0.8303(11 - 8, 5)$$

$$= -4.30535$$

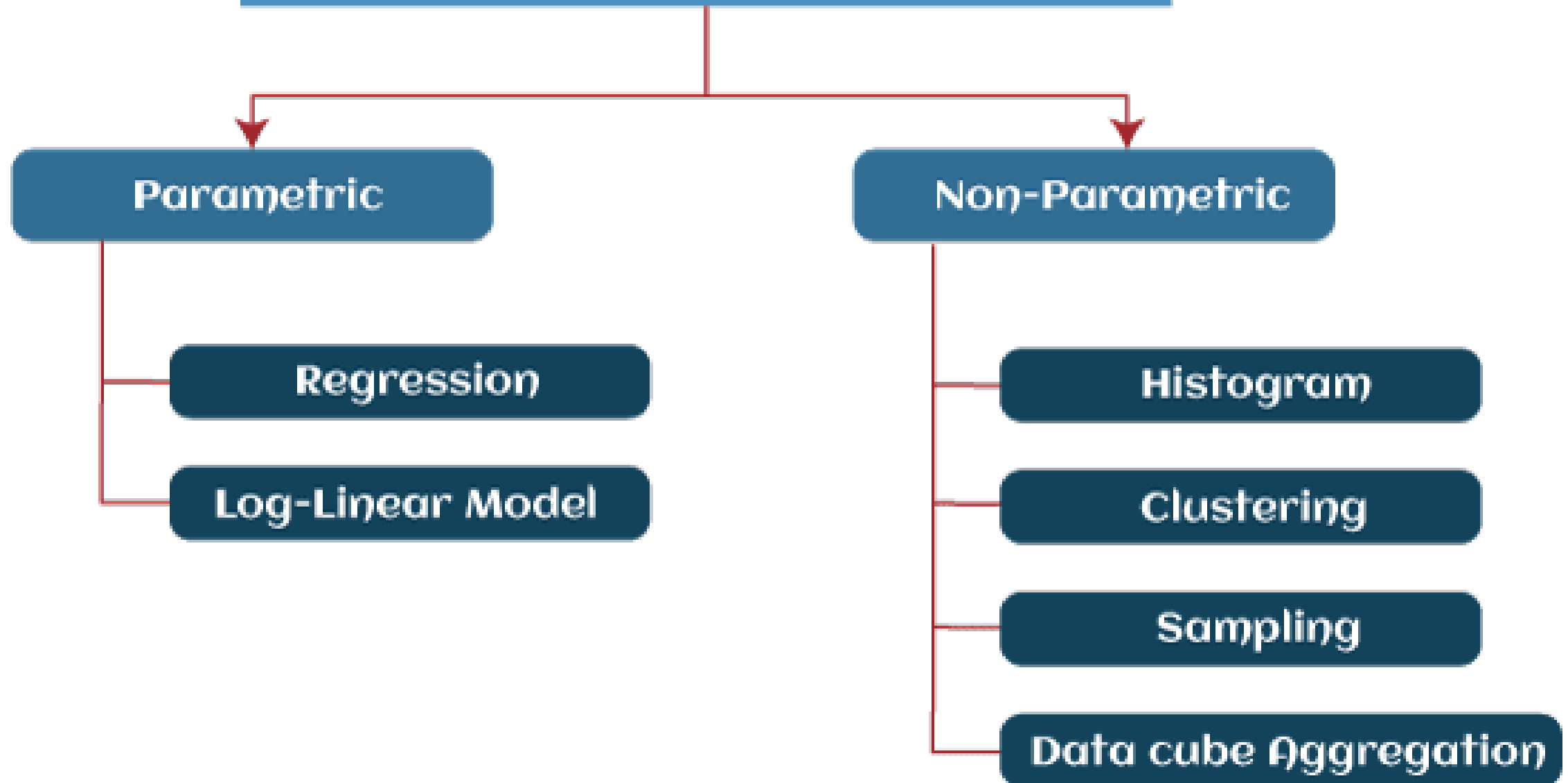
Feature	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14
First Principle Components	-4.3052	3.7361	5.6928	-5.1238

EXAMPLE

Features	E_1	E_2	E_3	E_4
X_1	2	3	4	5
X_2	4	6	8	10

- **Numerosity Reduction** – Data is replaced by estimated value / alter value
- Data replaced with mathematical models or smaller form of representation of the data instead of actual data.
- Parametric
- Non-Parametric

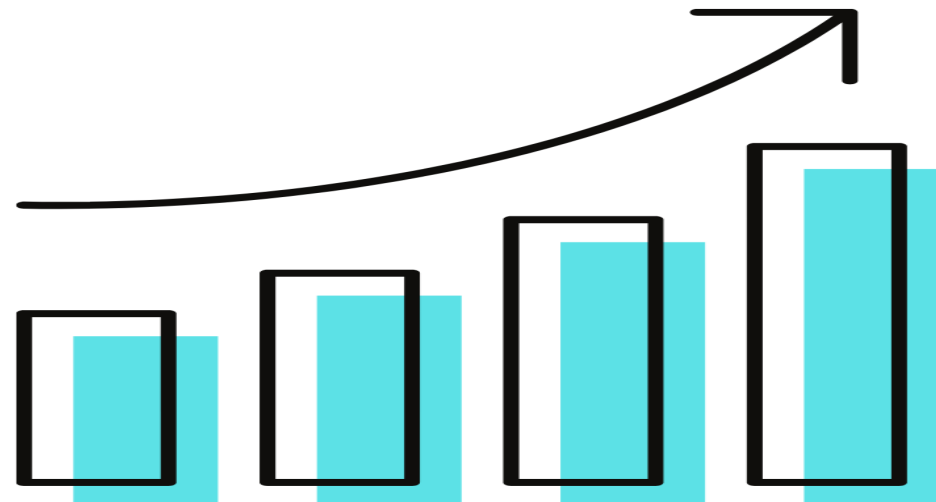
Types of Numerosity Reduction



- Parametric: model into which the data fits
- $y = wx + b$
- Non Parametric: Not using model

Histogram, Clustering, Sampling, Data Cube Aggregation, Data Compression.

- **Histograms**
- divisions the data distribution n of A into disjoint subsets, or buckets
- A histogram is a graph that represents frequency distribution which describes how often a value appears in the data



Histogram

- **Clustering:**
- Clustering techniques groups similar objects from the data so that the objects in a cluster are similar to each other, but they are dissimilar to objects in another cluster.
- The quality of the cluster depends on the **diameter of the cluster**, i.e., the max distance between any two objects in the cluster.



- **Sampling:** One of the methods used for data reduction is sampling, as it can reduce the large data set into a much smaller data sample

condt

- Data Cube Aggregation:
- It is a process in which info is gathered and expressed in summary form
- Dataset represented in smaller volume

condt

- **Data Compression** –Encoding of data
 - Lossless data reduction –compressed data can reconstructed to form original data without losing any information
 - Loss data reduction- Unable to reconstruct the original data from compressed one

PCA(Principal Component Analysis)

- Step 1: data set 1:

Example 1	Example 2	Example 3	Example 4	exampleN
X1	X11	X21	X31	X1N
X2	X12	X22	X32	XnN

- Step 2: mean of variable

$$\bar{X}_i = 1 / N (X_{i1} + X_{i2} + \dots + X_{iN})$$

- Step 3: Calculate covariance matrix

Covariance of all Pair (X_i, X_j)

$$\text{COV}(X_i, X_j) = \frac{1}{N-1} \sum_{k=1}^N (X_{ik} - \bar{X}_i)(X_{jk} - \bar{X}_j)$$

condt

- Construct N* N Covariance Matrix

$$S = \begin{matrix} & \begin{matrix} cov(x1,x1) & cov(x1,x2) & cov(x1,xn) \end{matrix} \\ \begin{matrix} cov(x2,x1) & cov(x2,x2) & cov(x2,xn) \\ cov(xn,x1) & cov(xn,x2) & cov(xn,xn) \end{matrix} & \end{matrix}$$

- Find eigen values and vector
- Normalized eigen values and vector
- Derive New Dataset

Example

- Given Following Data use PCA to reduce the dimension from 2 to 1

Features	Ex1	Ex2	Ex3	Ex4
X	4	8	13	7
Y	11	4	5	14

- Step1:
- No of Features: $n=2$
- No. of sample: $N=4$

condt

Step 2:

$$\text{Mean}(X_j = 4 + 8 + 13 + 7)/4 = 8$$

$$\text{Mean}(Y_j = 11 + 4 + 5 + 14)/4 = 8.5$$

Step 3:

Computation of covariance Matrix

$$(x,y) = (x,x)(x,y)(y,x)(y,y)$$

$$\text{Cov}(x,x) = 1/(N-1) \sum_{k=1}^N (x_i - \bar{X})^2$$

Introduction To Data Warehousing

- Database-Collection of related data
 - structured with defined schema
 - Information organized in rows and columns
 - Ex MySQL, ORACLE, SQL
- Data Warehouse- Special type of database
 - Analysis and report
 - Support for Business Intelligence Activities
 - Ex:How Much % percentage Discount
 - MY SQL with Special Data warehouse

Condt

- Data Lake- It is centralized repository that allows you to store all structured and unstructured data at any scale
- Ex: AWS, Azure, BigQuery...

Database Vs Datawarehouse

S.No	Database	Datawarehouse
1.	Transactional Based	Analytical Based
2.	Used to Store data ,retrieve,..	Used to Analysis data Prediction of data Decision support
3.	OLTP	OLAP
4.	Fundamental operation of business	Help to analyse the business
5.	Day to Day Processing	Historical Processing
6.	Relational Database	Multidimensional arrays

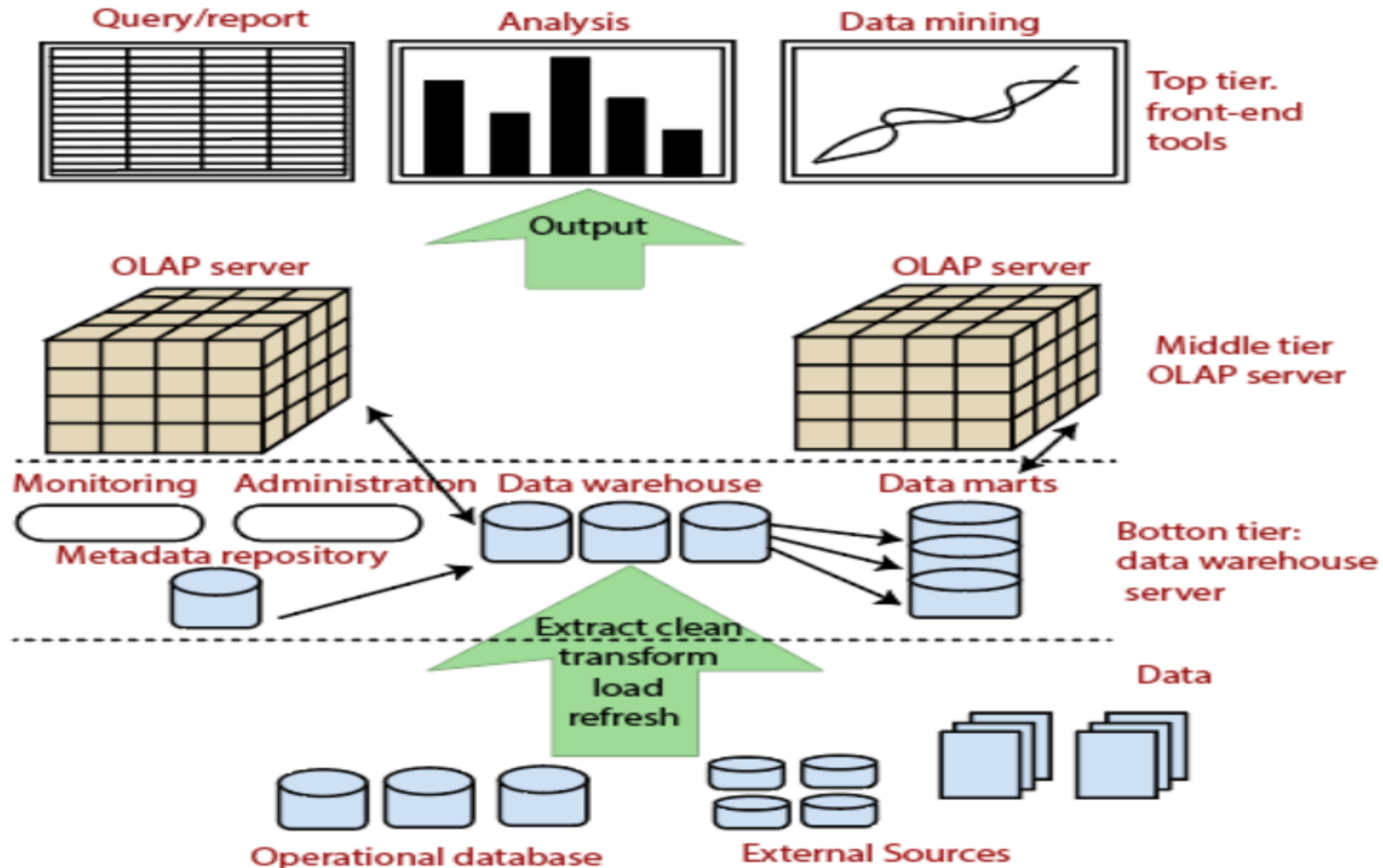
Features of DW

- Subjected oriented-It provides information catered to a **specific subject** instead of the **whole organization's ongoing operations**
- Integrated-it is developed by **combining data from multiple sources**, such as flat files and relational databases, which offers better data analysis.
- Time variant -The data in a DWH gives information from a **specific historical point of time**. The data is categorized with a particular time frame.
- Non –volatile -Non-volatile refers to **historical data that is not omitted when newer data is added**.
 - previous data is not erased when new data is entered in it

Three-Tier Data Warehouse Architecture

- Three layer/Tier
 - 1.Bottom Tier (Data Warehouse Server)
 - 2.Middle Tier (OLAP Server)
 - 3.Top Tier (Front end Tools).

Architecture:



Three-Tier Data Warehouse Architecture

- Database of the DW server
- It is usually relational DBMS
- Fetch only relevant information based on data mining request .
- Data is cleaned , transformed and loaded into this layer using back end tools (ETL)
- Function provided by back end tools are:

Data extraction

Data cleaning

Data transformation

Data load

Data refresh

Component in Data Warehouse

1. Data Warehouse
2. Meta Data-data about data , defining DW
 - Used for building and maintaining and managing DW
 - Includes : Data name, definition, source of data, algorithm used for cleaning

For Example: 103 NK78 30000

Model No:103 ; NK78-Agent ID no ; Total sales-30000;

- Technical Meta Data- contains **information** about **DW**
 - used by **database designer and administrators**
- Business Meta Data-contains information about data ,
 - used by end user
 - without meta data we cant conclude , what its represent

3.Data Mart:- Subsidiary of data warehouse ,contain small slice of data

Example: single department data

Dependent :sourced directly from DW

Independent: Sourced from one or more database

4.Monitoring and administration:

- Includes refreshment and data recovery
- Limiting the size of DW controlling the number and ranges of queries, manage the data base performance

- Middle Tier: OLAP server
- Concept of multidimensional database
- It allows users to analyze the data using multidimensional views
- OLAP Server Implemented by two models:
ROLAP
MOLAP

OLAP SERVER TYPES

- ROLAP
- MOLAP
- HOLAP

- ROLAP:
 - Relational OLAP
 - Tools can analyze large amount of data
 - ROLAP tools store and analyze volatile data
 - Expertise person can deal with ROLAP
 - Dis adv: Poor query performance

condt

- MOLAP:

- Multidimensional OLAP
- MOLAP easy to use
- Information retrieval is fast
- Can perform complex computation

HOLAP:

- Hybrid OLAP
- Complex computation
- large amount data

Condt

- Top Tier: Front ends Tools/ Front End client Layer
- **Query tool and report tool:** production of report
- Query tool: creation of SQL
- Application dependent tool:
- Data mining Tool:

Multidimensional Data Model

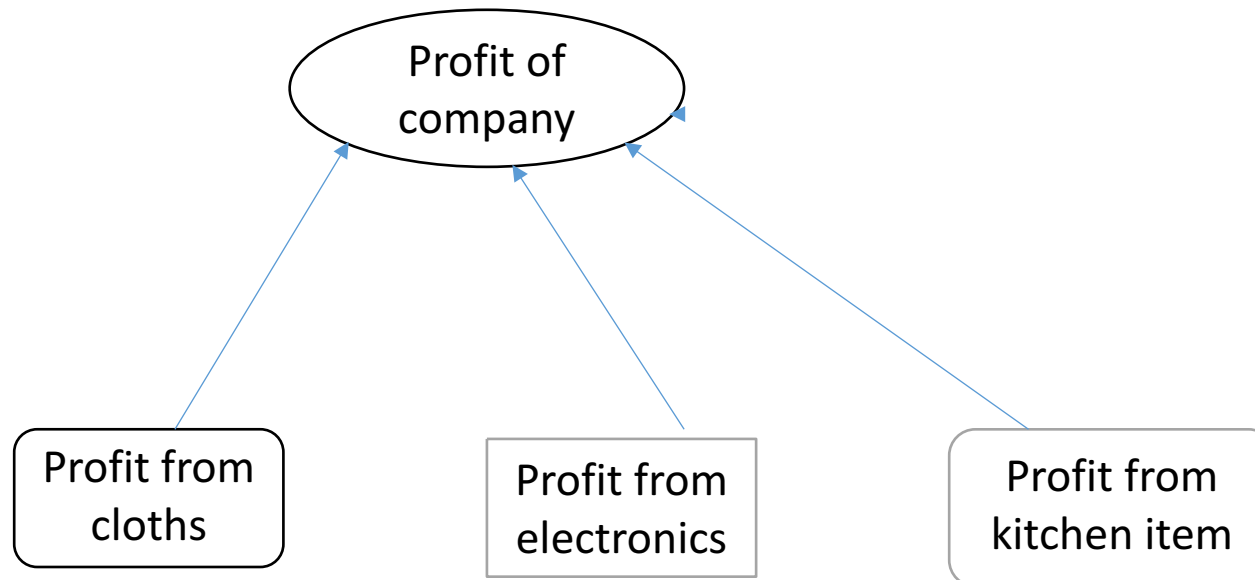
OLAP Cube –Data structure used quick data analysis

ETL tools- Data's loaded into OLAP Cube

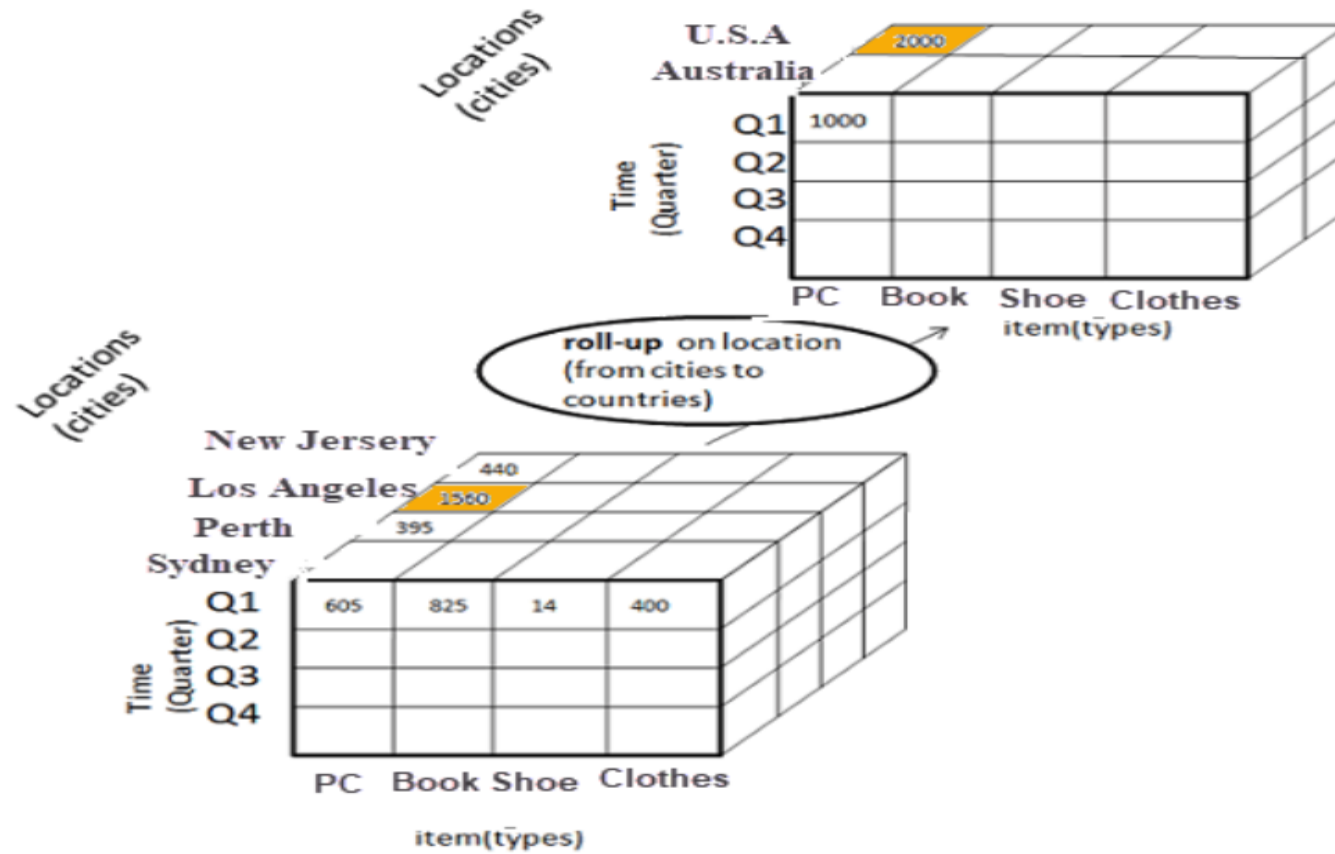
4 operations:

1.Roll-up(drill –up)

- Uses the concept of Hierarchy

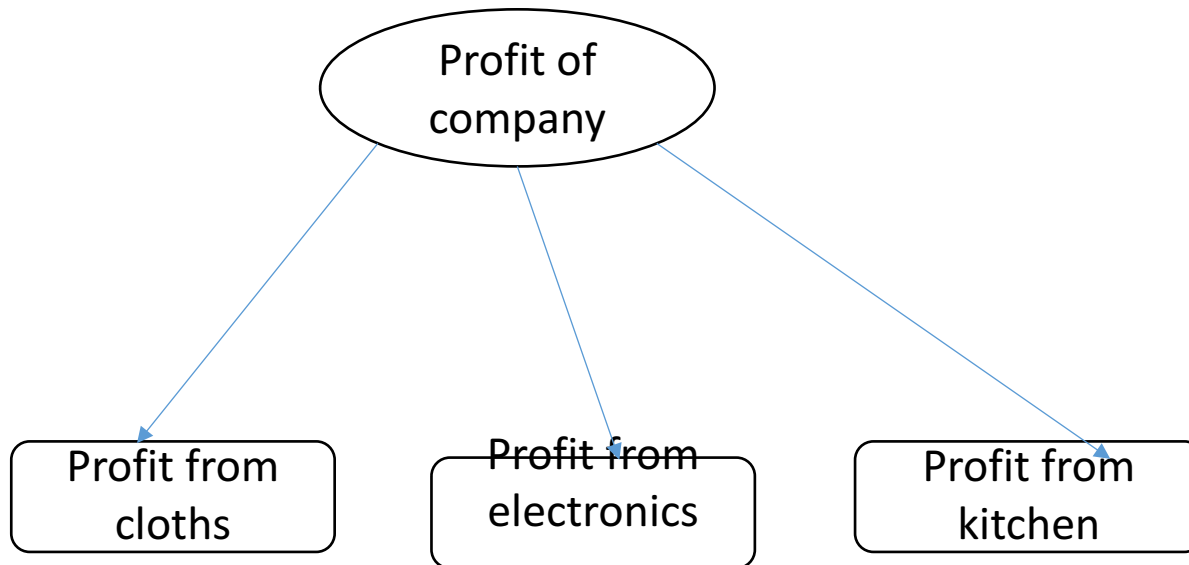


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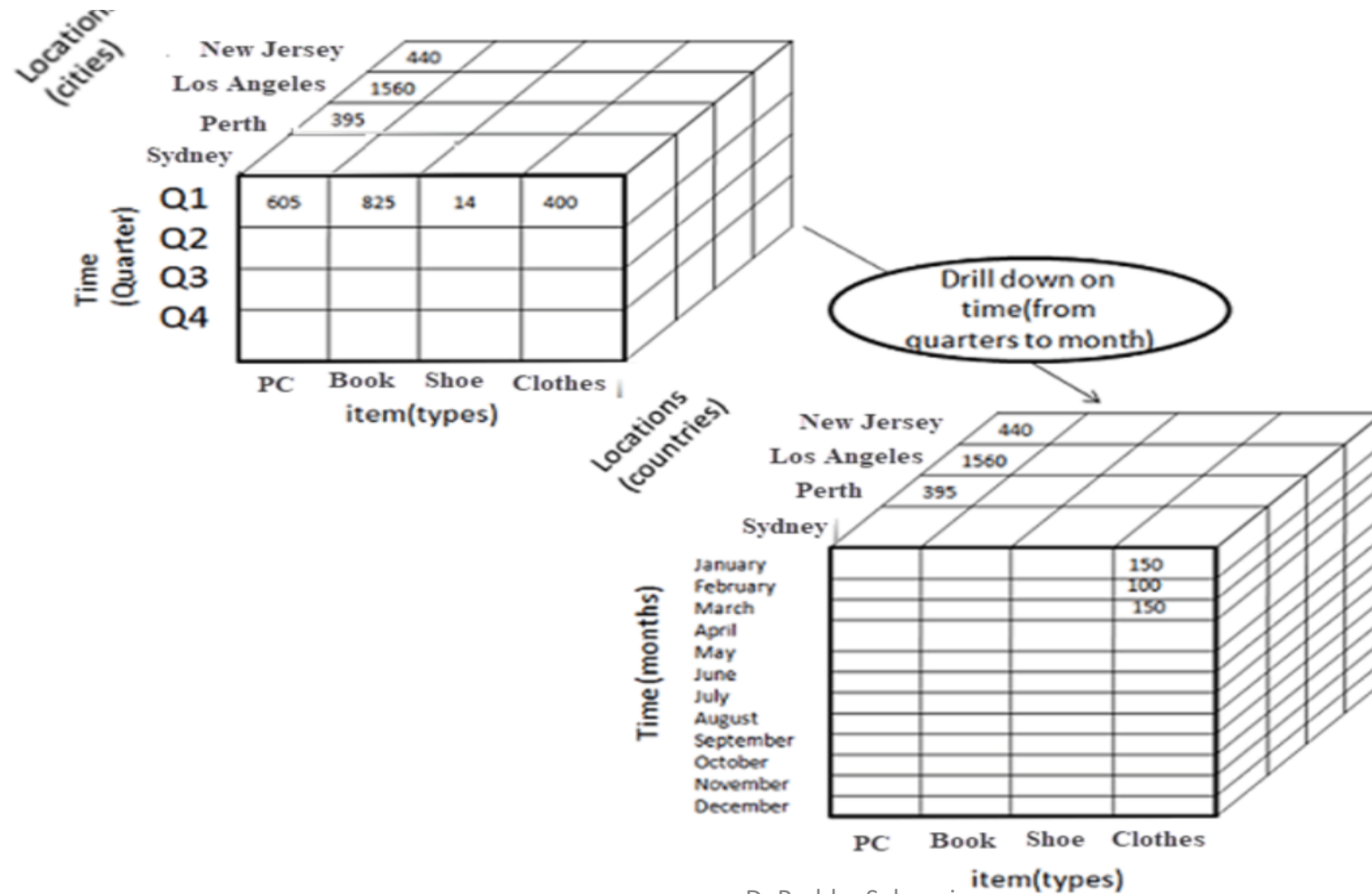


condt

- Drill Down-
- -Reverse of drill up
- - data is drilled down



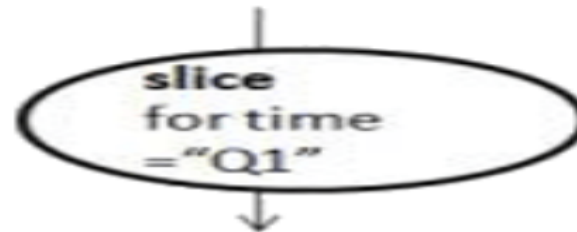
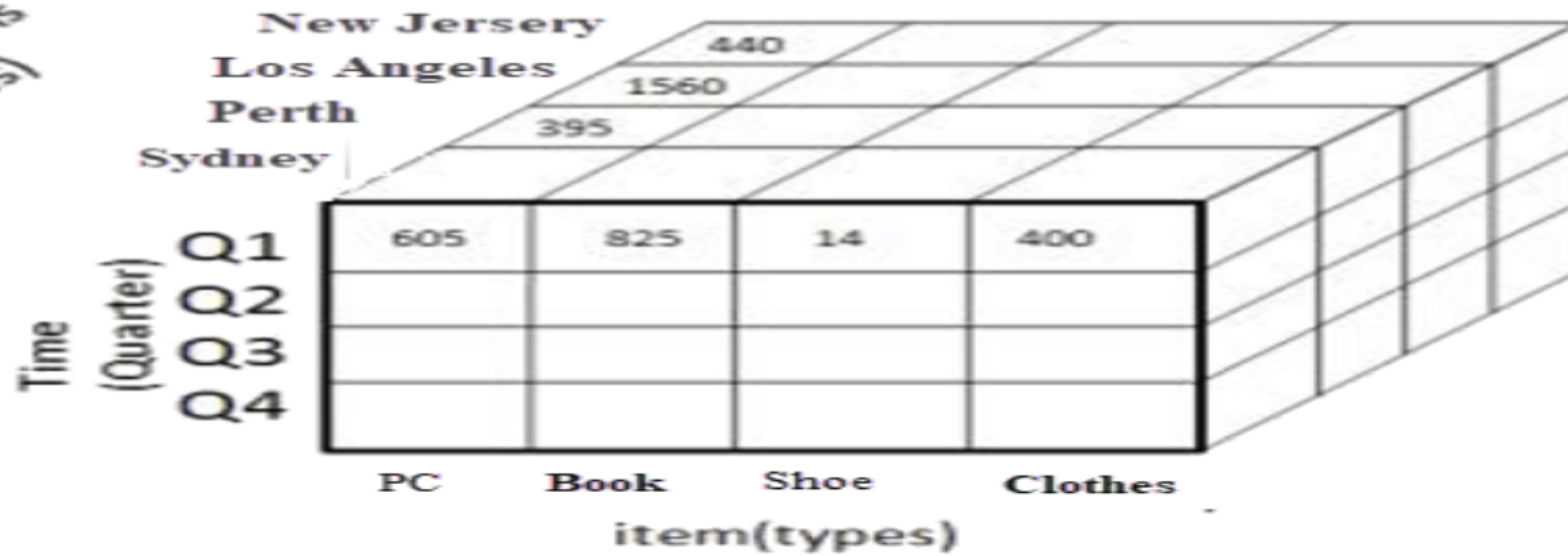
condt



Condt

- Slice :
- Data cube is sliced and information is divided and new cube's is formed
- Dimension Time is Sliced with Q1 as the filter.
- A new cube is created altogether.

Locations
(cities)



cities

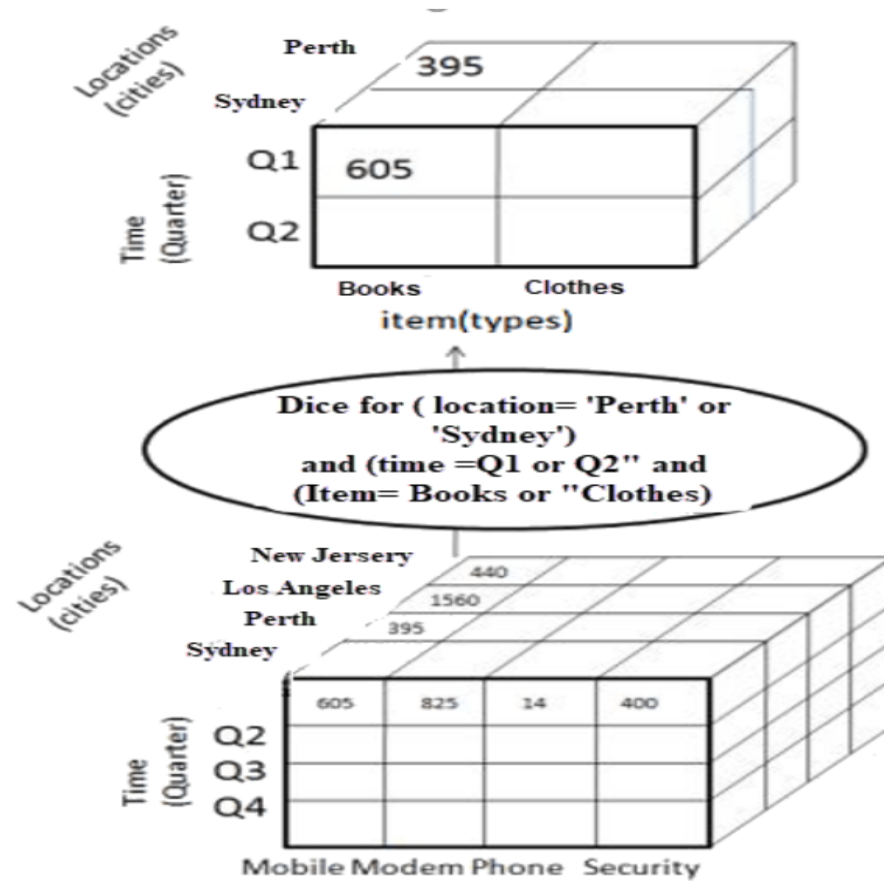
New Jersey				
Los Angeles				
Perth				
Sydney	605	825	14	400
	PC	Book	Shoe	Clothes

item(types)

Condt

- **Dice:**
- This operation is similar to a slice. The difference in dice is you select 2 or more dimensions that result in the creation of a sub-cube.

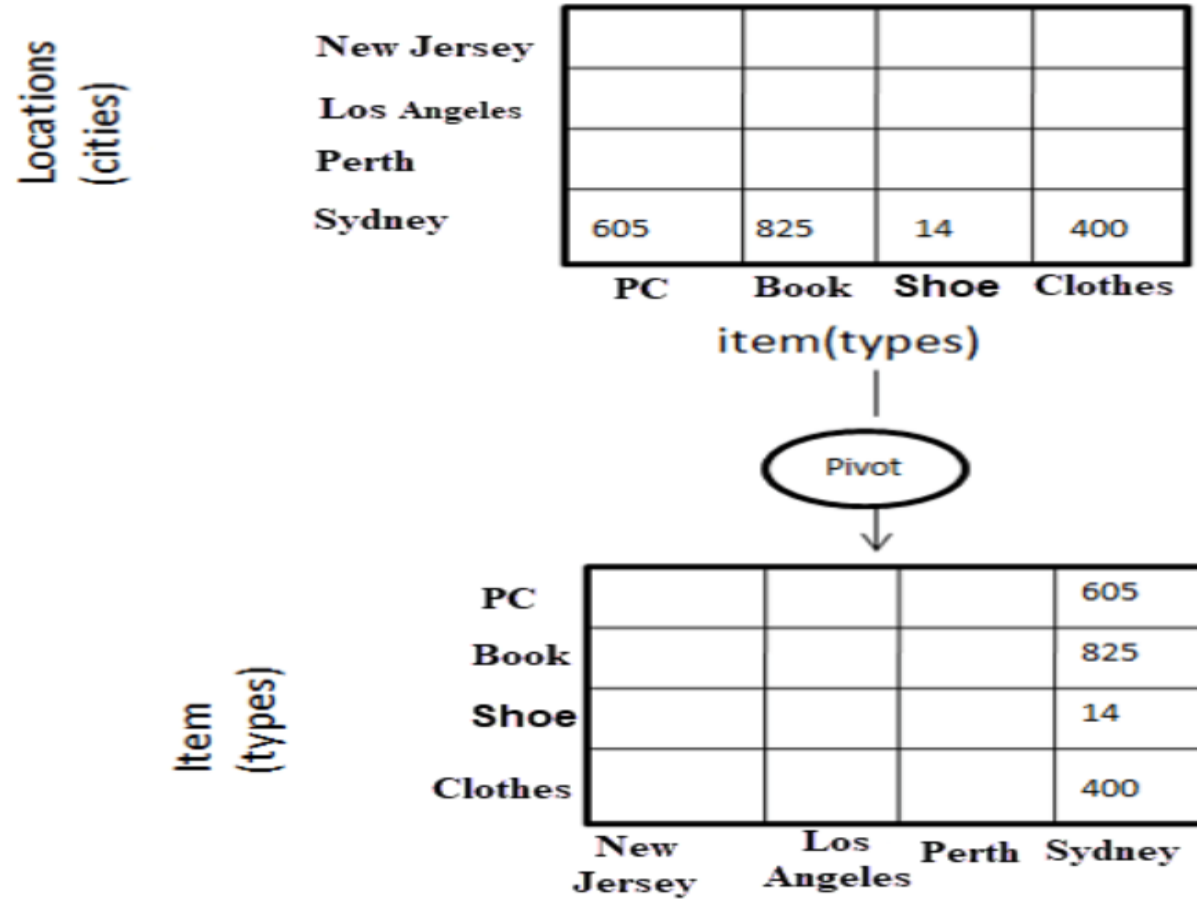
condt



Condt

- **4) Pivot**
- In Pivot, you rotate the data axes to provide a substitute presentation of data.
- In the following example, the pivot is based on item types.

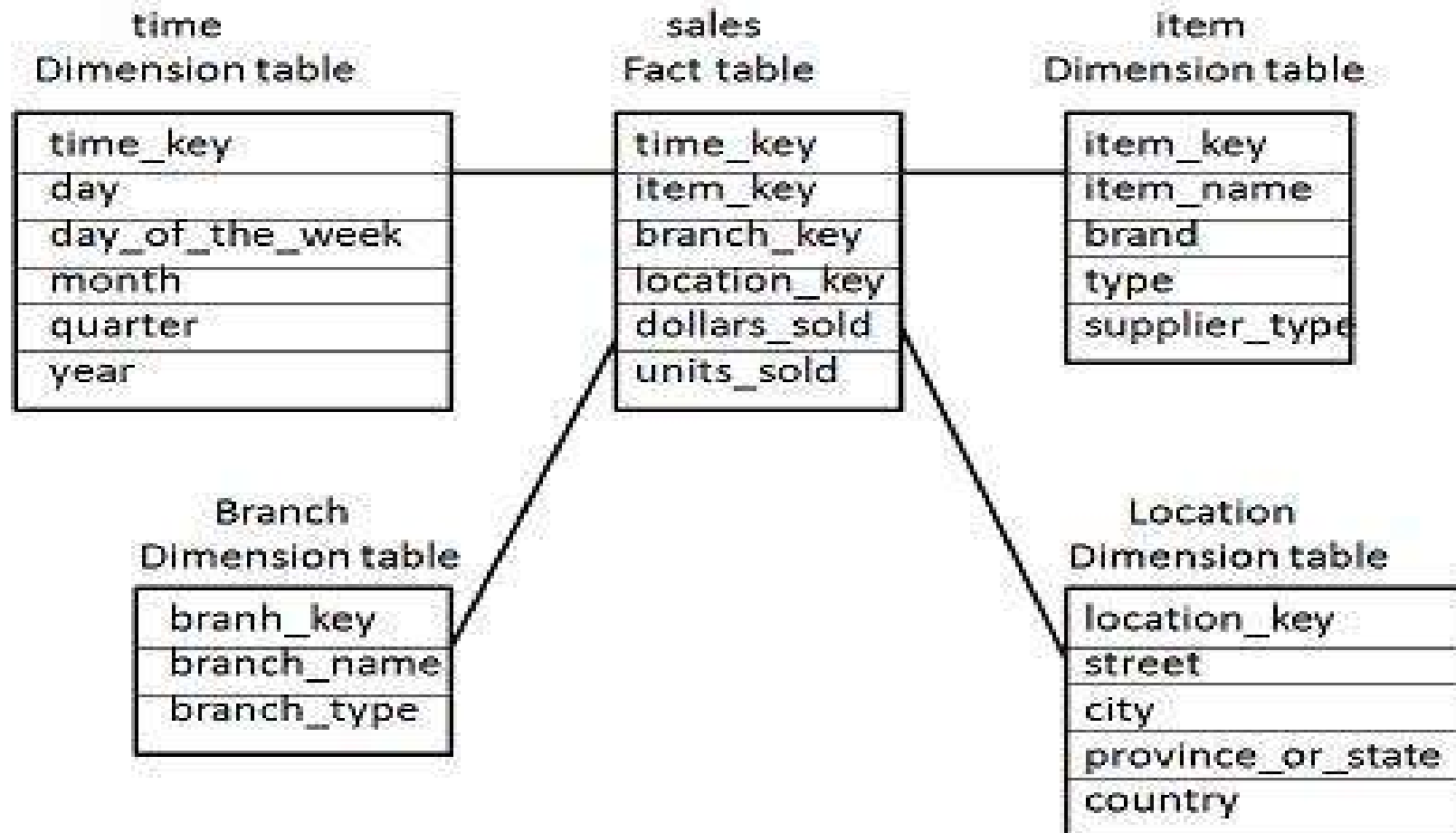
condt



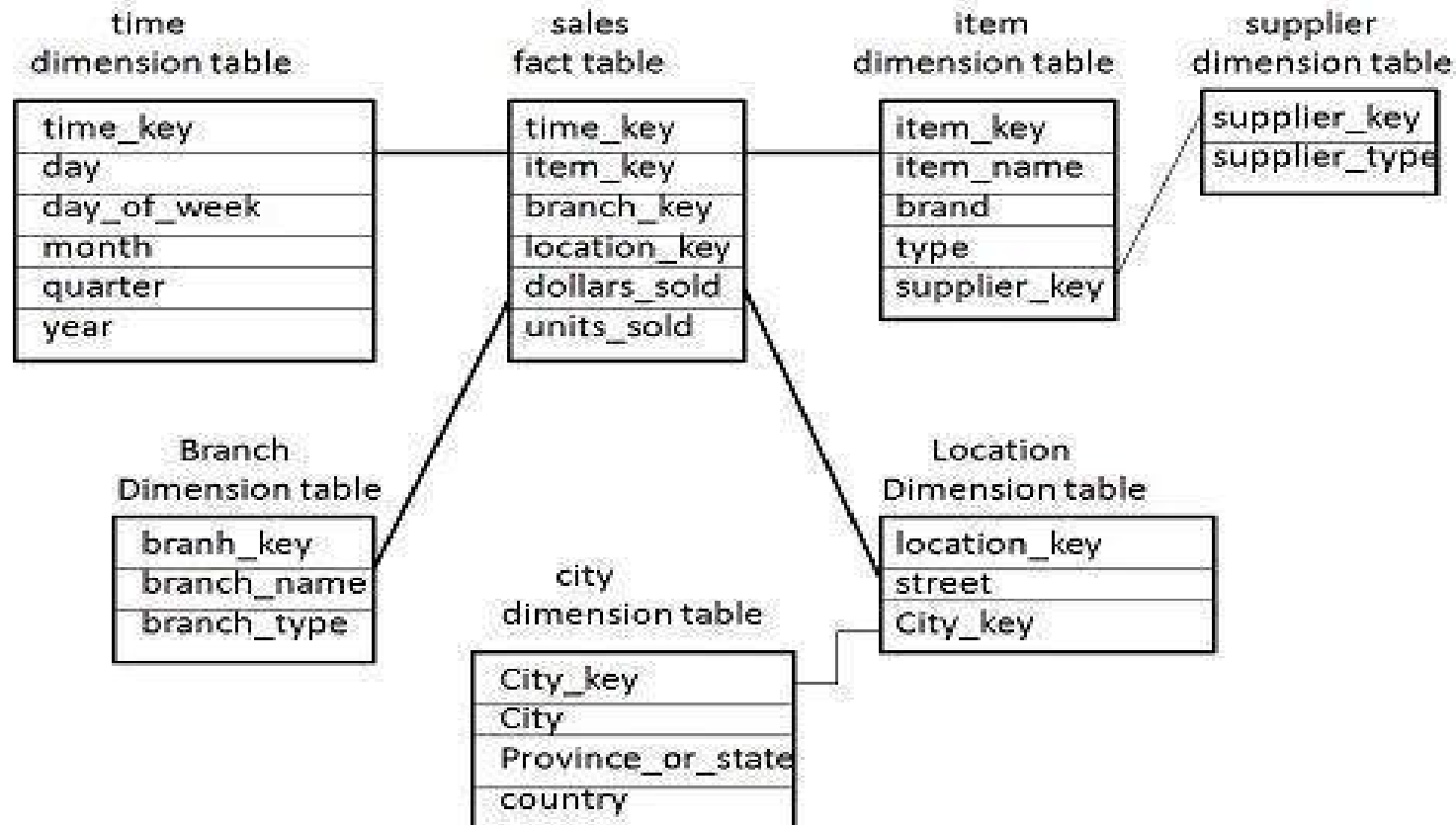
Schema in DW

- Schema is a logical description of the entire database.
- **Star Schema**
- **Snowflake Schema**
- **Fact Schema**
- **Fact Table-Main Table**
- **Dimension Table**

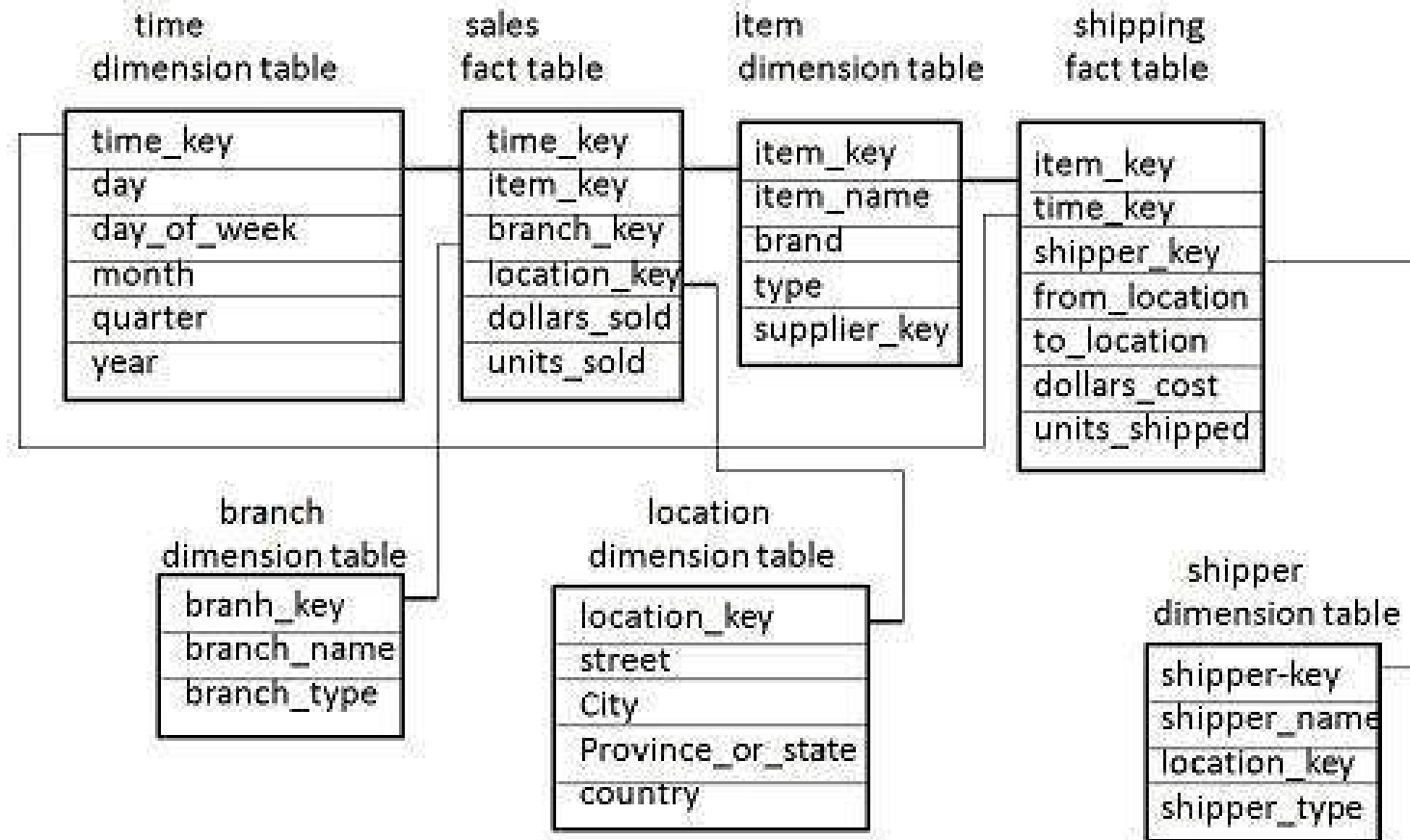
Star Schema:



Snowflake Schema



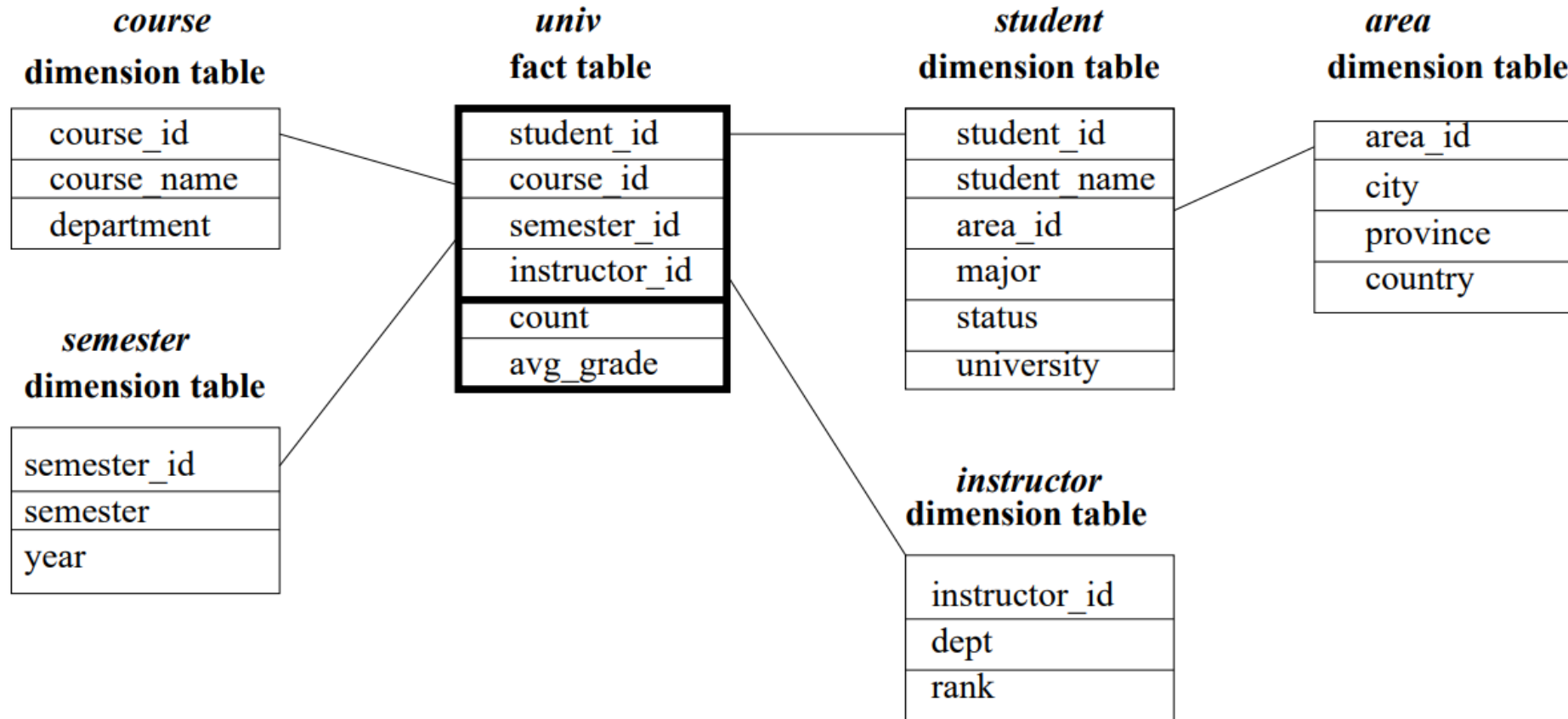
Fact/Galaxy



Ex:01

- Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (a) Draw a snowflake schema diagram for the data warehouse.

Answer:



Ex:No:02

- Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit

Answer

